



BIOLUMOS

INTRODUCTION TO THE INNOVATION



INTRODUCTION

A BIT OF HISTORY



Light therapy is one of the oldest non-pharmacological treatment methods. It originated with heliotherapy, and with the advent of artificial light sources, it became widely used.

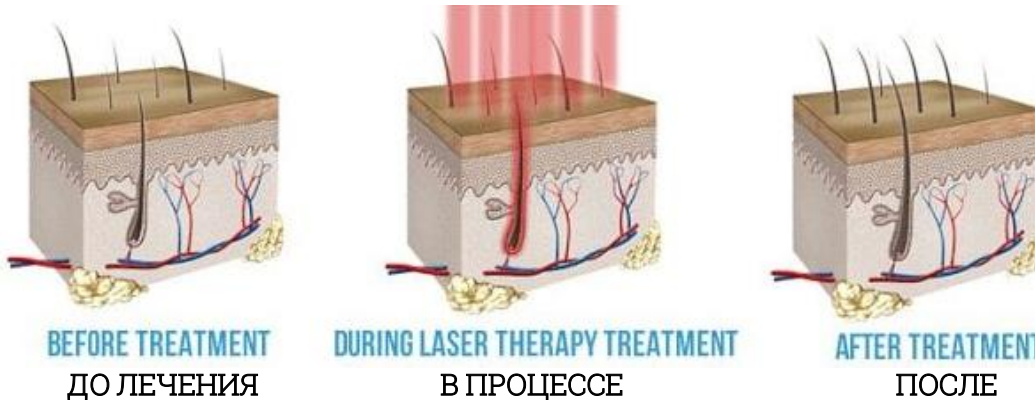


PHOTOBIOMODULATION

FIRST MENTIONS

This is a change in the functional activity of cells under the influence of light.

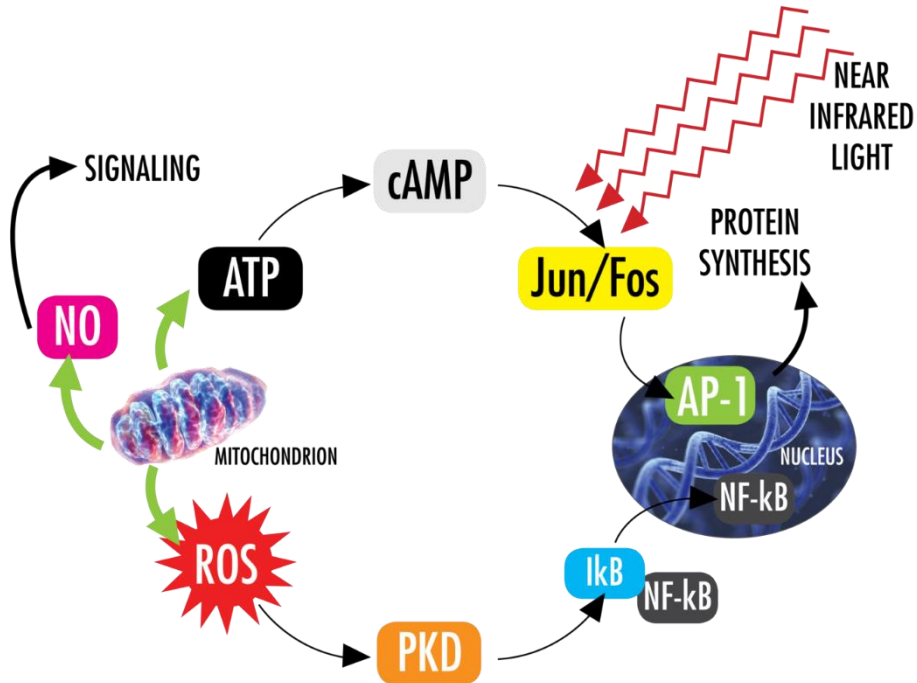
In experiments on mice, he observed accelerated hair growth after exposure to light radiation.



Endre Mester

MECHANISM OF ACTION

KEY DEFINITIONS



Light exposure affects three key components:

- Absorption of light radiation by mitochondria with conversion into ATP (is the cell's universal energy molecule, providing fuel for essential biological processes.);
- The creation of reactive oxygen species, which promotes gene transcription and subsequent cellular repair;
- The release of nitric oxide, which contributes to vasodilation and increases oxygen content in the blood.

ATP (Adenosine triphosphate)

cAMP (Cyclic adenosine monophosphate)

Jun/Fos (proteins, components of the AP-1 transcription factor)

AP-1 (Activator protein 1, enhances gene transcription)

ROS (Reactive oxygen species)

PKD (Protein kinase D)

IκB (Transcription factor) + **NF-κB** (signaling pathway)

NF-κB (Stimulates gene transcription)

NO (Nitric oxide)

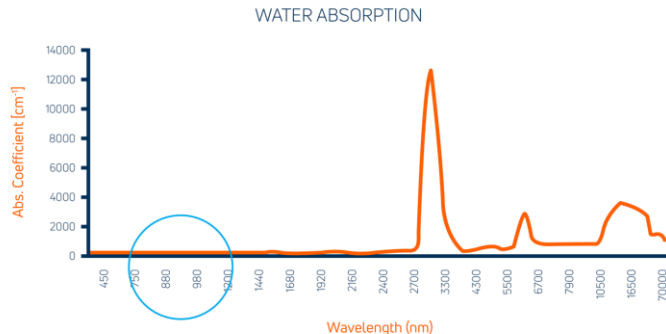
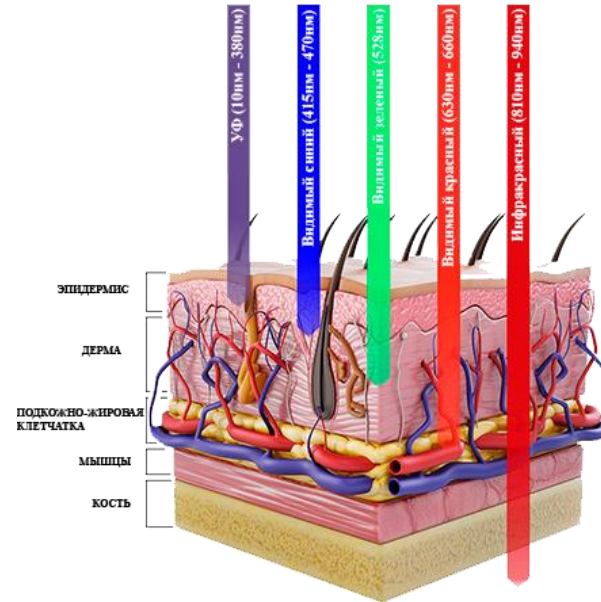
<https://lightforcemedical.com/photobiomodulation-therapy-pbm/>

<https://www.vielight.com/photobiomodulation-science/>

DOI:10.1089/photob.2022.0127

PHOTOBIOMODULATION

CHOICE OF LED TYPE



The 810 nm wavelength is the most effective for medical purposes. This is explained by the fact that the light beam can penetrate deep through bone tissue—up to 4 inches.

<https://lightforcemedical.com/photobiomodulation-therapy-pbm/>
<https://www.vielight.com/photobiomodulation-science/>
DOI: 10.1016/B978-0-12-815305-5.00005-1

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SAFETY

According to data from the literature, confirmed in both animal and human studies, photobiomodulation is considered a **safe technology**.

Power levels below **100 mW/cm²** are regarded as safe because the temperature increase does not exceed **32.68°F (0.38°C)** in the scalp and **32.11°F (0.06°C)** in the brain.

Paolillo F.R., Luccas G.A.A., Parizotto N.A., Paolillo A.R., De Castro Neto J.C., Bagnato V.S. The effects of transcranial laser photobiomodulation and neuromuscular electrical stimulation in the treatment of post-stroke dysfunctions//Journal of Biophotonics, 2023, Vol. 16, No. 4, P. e202200260.

Iosifescu D., Collins K., Hurtado-Puerto A., Irvin M., Clancy J., Sparpana A., Sullivan E., Parincu Z., Ratai E.-M., Funes C., Weerasekera A., Dmochowski J., Cassano P. Grant Report on the Transcranial near Infrared Radiation and Cerebral Blood Flow in Depression (TRIADE) Study//Photonics, 2023, Vol. 10, No. 1, P. 90.

Guillen A.R., Truong D.Q., Faria P.C., Pryor B., De Taboada L., Datta A. High-resolution computational modeling of Transcranial Photobiomodulation: Light propagation and thermal effects. High-resolution computational modeling of Transcranial Photobiomodulation. - 2025.

Hacke W., Schellinger P.D., Albers G.W., Bornstein N.M., Dahlof B.L., Segura T., et al. Transcranial Laser Therapy in Acute Stroke Treatment: Results of Neurothera Effectiveness and Safety Trial 3, a Phase III Clinical End Point Device Trial//Stroke, 2014, Vol. 45, Transcranial Laser Therapy in Acute Stroke Treatment, No. 11, P. 3187-3193.

PHOTOBIOIMODULATION

APPLICATION IN SPORTS

REHABILITATIVE MEDICINE

PERFORMANCE ENHANCEMENT

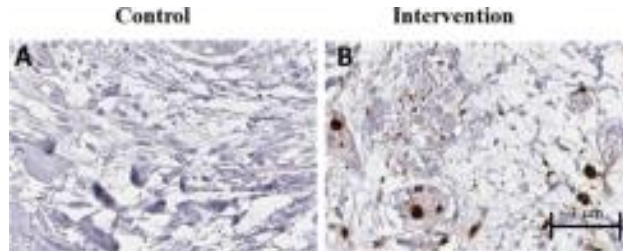


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SPORTS: EFFECTS

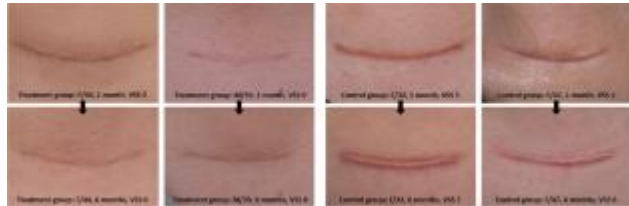
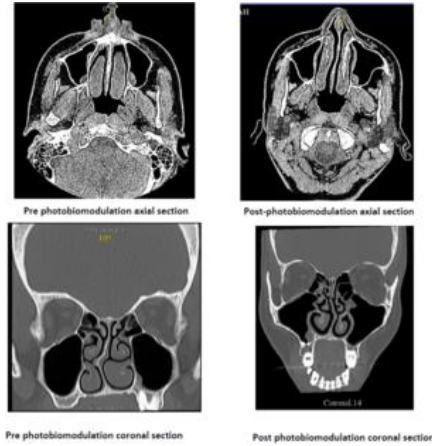
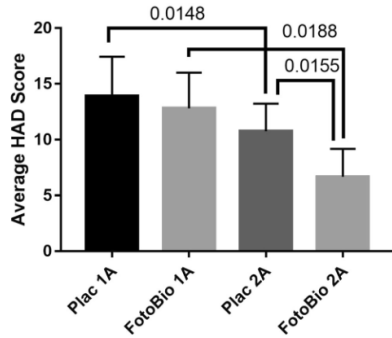


- Antioxidant effect, reducing oxidative stress caused by physical activity
- Improved aerobic endurance and faster recovery after exercise
- Lipolysis and apoptosis of adipose tissue (“fat burning”)



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REHABILITATIVE MEDICINE



- Reduction of headaches and fatigue
- Pain relief
- Prevention of hypertrophic scar formation
- Improvement of brain activity
- Reduction of anxiety and depression
- And much more

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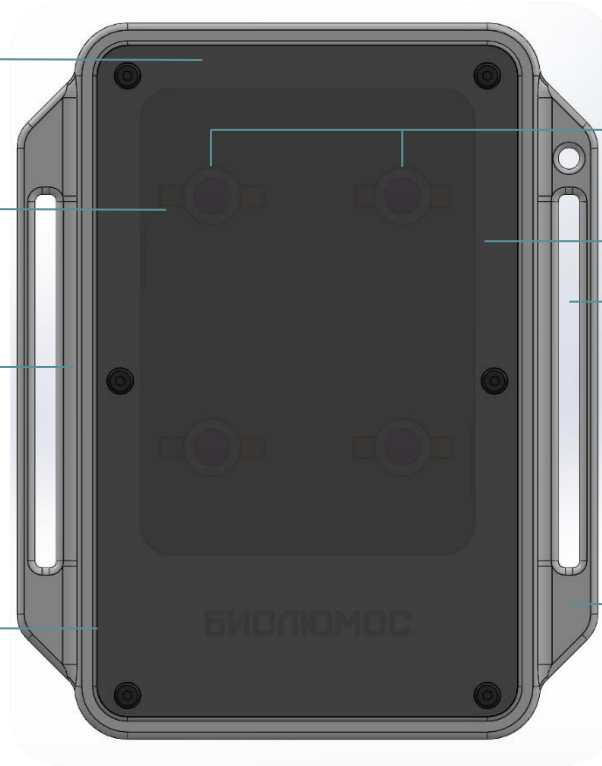
DETAILS

Wi-Fi and
Bluetooth

Dual-core
processor

Charging and
firmware updates
via Type-C port

Built-in Li-ion
battery



4 high-power LEDs
(808–810 nm)

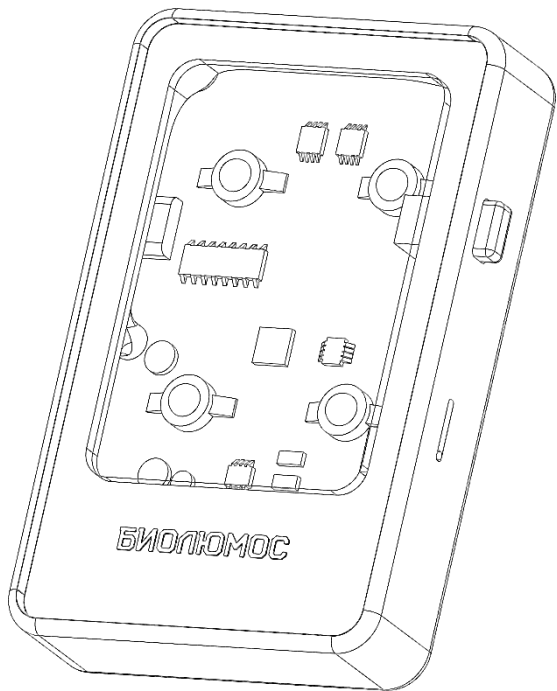
Transparent acrylic

Strap loop for
attachment

Soft thermoplastic
polyurethane (TPU)
case

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ADVANTAGES



High effectiveness	Uses 4 high-power LEDs with a wavelength of 808–810 nm
Compact size	Conveniently fits in the hand (dimensions: W 2.46 × H 3.84 × D 0.71 inches)
Versatility	Adjustable flashing frequency from 1 Hz to 10 kHz
Ease of use	Operation modes can be controlled and configured via a mobile phone
Autonomy	No external power source required
Wide applicability	Can be applied locally to all areas of the body
Affordability	Device cost lower than comparable alternatives

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MOBILE APPLICATION

Compatibility	Android and iOS
Presets	The app includes 23 device exposure scenarios, developed with AI based on international scientific research
Simplicity	Minimal steps required to start using the device
Convenience	Countdown timer, battery level, notifications, and other functions
Quick connection	Automatically connects to the last used session
Versatility	Ability to manually set any exposure mode

● Подключено
BiolumosBT_07:64

Отключить

Информация об устройстве

Уровень заряда:
84%

Температура:
40°C

Время работы:
1ч 16м

Состояние:
Активен

Активный курсВ процессе

Универсальный

Прогресс: 0 из 10 сеансов **0%**

• 5 раз в неделю

• 20 мин на сеанс

• Следующий сеанс: 11.08.2025

Остановить

Завершить курс

Активный сеанс

19:30

Параметры управления

Главная

Пресеты

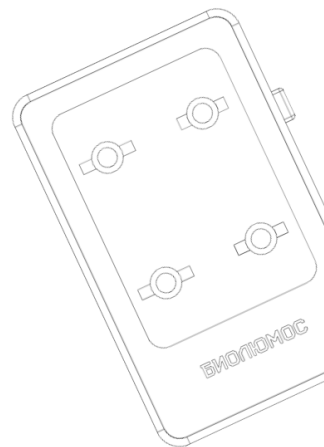
История

Настройки

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MANUAL

РУКОВОДСТВО ПО ЭКСПЛУАТАЦИИ



БИОЛЮМОС УЛЬТРА

2

В ваших руках путь к
технологии будущего

Начало

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CONCLUSION



BIOLUMOS is the most advanced and effective device for treatment and rehabilitation across a range of conditions.

The technology is also a promising tool for achieving high-level athletic performance.



THANK YOU FOR
YOUR ATTENTION



Telegram: @biolumos_news